Devops Syllabus

Topic Cover

1. Ansible Automation
2. Docker Container
3. Kubernetes
4. Terraform
5. Git / Github
6. Jenkins
7. Monitoring Tool (Nagios , Prometheus , Grafana)

Ansible Automation :-

i) What is Ansible?
ii) Ansible Lab setup
iii) Inventory File entry
iv) Make custom Inventory file and separate Inventory file
v) Use of Modules (Ad-Hoc Method)
vi) Playbook (Dictionary, Single line, Multiline Format)
vii) Playbook Advance
viii) Variables
ix) Loop
x) Fact
xi) Condition statement
xii) Notify and Handler
xiii) Tags
xiv) Ignore error
xv) Rescue, Block and always
xvi) Ansible Vault
xvii) Handle windows client from ansible server
xviii) Role Structure
xix) Ansible Roles Galaxy
xx) Ansible use in AWS
xxi) Ansible AWS Live project

Docker Container

i) What is Docker and Docker container
ii) Docker basic command
iii) Docker create, Push, & Pull a Docker Image from Docker Hub
iv) Docker Volumes, Mount & Docker Storage
v) Host your own docker registry
vi) Setup Private Docker Registry- Secure SSL Certificate
vii) Docker Swarm
viii) Docker file
ix) Docker Compose
x) Docker Network
xi) Ansible use in Docker
xii) Ansible Docker Live Project

Kubernetes

i) Kubernetes Introduction
ii) Single server node cluster configuration (minikube)
iii) Kubernetes Cluster using Kubeadm
iv) Kubernetes Cluster Controlling from Windows OS
v) Kubernetes Cluster Controlling Linux OS
vi) Kubernetes Kubeadm Cluster Upgrade from Older Version to New Version
vii) Kubernetes Cluster Reset
viii) Kubernetes Cluster Token & CA Public Key Use Case
ix) Kubernetes How to Create Pod
x) Kubernetes Replication Controller
xi) Kubernetes Replication Controller Rolling-Update
xii) Kubernetes Deployment Lab
xiii) Kubernetes Deployment Rollout, Rollback, pause & resume
xiv) Kubernetes Rolling Deployment & Recreate LAB
xv) Kubernetes Services Explained
xvi) Kubernetes Services Cluster IP
xvii) Kubernetes Services NodePort LAB
xviii) Kubernetes Services Load Balancer LAB
xix) Kubernetes Init Containers POD Example
xx) Kubernetes How to create Multi-Container POD LAB
xxi) Setup Kubernetes Cluster On AWS
xxii) Kubernetes Kubeless Cluster Use Case
xxiii) Kubernetes Replica Set
xxiv) How to create Multi-Container POD
xxv) Kubernetes Persistent Volume (PV) and Persistent Volume Claim (PVC)
xxvi) Kubernetes Namespace
xxvii) Kubernetes Taint
xxviii) Kubernetes Labels
xxix) How To Set Pods Limits
xxx) Kubernetes Resources Quota
xxxi) Kubernetes Limit Range
xxxii) Horizontal Pod Auto-Scaling

Terraform

i) Terraform Introduction
ii) Terraform use in AWS cloud and make EC2, EBS, VPC
iii) Terraform live project in AWS cloud
iv) Terraform variable
v) Count parameter
vi) Conditions
vii) Local value
viii) Output value
ix) Terraform State Management, Shared Storage for State File, Remote Backend
x) Terraform provisioners

xi) Packer Build Automated Machine Image
xii) Packer Lab
xiii) Packer Authentication Methods
xiv) Packer Provisioner
xv) Packer Shell Provisioner with File Provisioner Live Practical Demo
xvi) Packer Ansible Provisioner with Project
xvii) Packer Ansible Local Provisioner
xviii) Packer Timestamp function Use
xix) Packer How to Use Multiple Builders
xx) packer override parameter
xxi) Packer pauses before parameter
xxii) Packer max_ retries parameter
xxiii) Packer timeout parameter

Git/GitHub

i) Introduction to Version Control System, GIT & GITHUB
ii) GIT & GITHUB Workflow
iii) Setup On-Premise GIT-Server
iv) GIT Command IN HINDI
v) Resolving Merge Conflict Error
vi) GIT Log Command
vii) GIT Branch
viii) GIT reset and reset Type
ix) GIT stash tag
x) How to create a GitHub account and work

Jenkins

i) Jenkins Introduction
ii) Jenkins Installation in Linux
iii) Jenkins Installation in Windows
iv) Jenkins Installation in Docker Container
v) How to Integrate GitHub with Jenkins
vi) DevOps Project With Ansible Part 1
vii) Devops Project with Docker
viii) Devops Project with Kubernetes
ix) CI / CD JOBS
x) Jenkins Pipelines with Groovy Script Scripted & Declarative
xi) Scripted Pipeline LAB With Real Industry Project Based

Nagios

i) Understand and appreciate Nagios as a Continuous Monitoring tool.
ii) Install Nagios
iii) Understand Nagios architecture and add a remote server to Nagios.
iv) Nagios Architecture

v) NRPE plugin
vi) Add NRPE to monitor using the server
vii) How to monitor the servers
viii) Add Service in monitoring

Prometheus

i) Prometheus architecture
ii) Time-series database
iii) Metrics, labels, samples
iv) Prometheus data model
v) Install Prometheus on Linux
vi) Prometheus file structure
vii) Prometheus service configuration
viii) prometheus.yml
ix) Global config
x) Scrape intervals
xi) Scrape configs
xii) Job and target concepts
xiii) Node Exporter
xiv) Process Exporter
xv) Blackbox Exporter
xvi) cAdvisor for Docker metrics
xvii) Custom Exporters (Python/Go basics)
xviii) Basic queries
xix) Rate / irate / increase functions
xx) Aggregations (sum, avg, max, min)
xxi) Vector matching
xxii) Recording rules
xxiii) Alerting rules
xxiv) Architecture
xxv) Install & configure Alertmanager
xxvi) Alert routing
xxvii) Silence, inhibit
xxviii) Email, Slack, Webhook alerts

Grafana

i) Install Grafana OSS
ii) Systemd configuration
iii) Add users & roles
iv) Create dashboard and panels
v) Time-series, Stat, Gauge, Heatmap, Bar chart panels
vi) Variables: query, interval, custom, constant
vii) Reusable dashboards
viii) Add Prometheus as datasource
ix) Configure connection

x) Test and troubleshoot
xi) Grafana alerting vs Prometheus alerting
xii) Alert rules, contact points
xiii) Notification policies
xiv) Email, Slack, Microsoft Teams alerts
xv) Templating
xvi) Dashboard JSON
xvii) Grafana provisioning (YAML-based config)
xviii) Linking dashboards
xix) Import famous dashboards from Grafana.com
xx) Connecting Prometheus to Grafana
xxi) Visualizing Prometheus data in Grafana
xxii) Building custom dashboards for:
xxiii) CPU, RAM, Disk, Load
xxiv) Network metrics
xxv) Docker & Kubernetes monitoring
xxvi) Database monitoring (MySQL, PostgreSQL)

Important Note

- Duration = 3.5 month
- Every tool has industry-based project
- E – Book and Interview question and answer provide
- After every class recorded video provide